

A Constructive Approach for Building Wavelet Bases in $L^2(\mathbb{R}^d, \mathbb{R}^m)$ with Optimal Properties

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Abstract

The main contribution of this paper is a constructive method for building separable multivariate vector-valued wavelet bases in the general framework of $L^2(\mathbb{R}^d, \mathbb{R}^m)$ for any $d, m \geq 1$. While separable wavelet bases in $L^2(\mathbb{R}^d, \mathbb{R})$ are well-established and widely applied, the explicit construction of truly vector-valued wavelet bases remains an open problem, even in the simplest case of $L^2(\mathbb{R}, \mathbb{R}^2)$, let alone in $L^2(\mathbb{R}^2, \mathbb{R}^2)$. In practice, the conventional approach applies standard separable wavelet bases of $L^2(\mathbb{R}^2, \mathbb{R})$ independently to each component of vector-valued signals in $L^2(\mathbb{R}^2, \mathbb{R}^2)$. However, this approach fails to capture the intrinsic vectorial structure of the signals. To address this limitation, we propose a constructive approach within the vector-valued wavelet framework, providing a systematic method for constructing such bases in the general case of $L^2(\mathbb{R}^d, \mathbb{R}^m)$. By linking m -multiwavelets to vector-valued wavelets, our approach not only enables the systematic construction of separable multivariate bases in $L^2(\mathbb{R}^d, \mathbb{R}^m)$ that satisfy the vector-valued multiresolution analysis but also ensures that these bases inherit key structural properties, making them well-suited for practical applications.

Key words : Multivariate wavelet bases; Vector-valued wavelet bases; Vector-valued multiresolution decomposition; m -multiwavelets; Separable wavelets.

1 Introduction

Since the 1990s, wavelets have gained significant attention in signal processing and data analysis due to their ability to capture both frequency and time-domain information simultaneously. This dual capability makes them particularly well-suited for analyzing non-stationary signals. Wavelets have become a standard theoretical framework for analyzing signals [2], processes [21], and multifractal functions [22]. A fundamental concept in this framework is multiresolution decomposition (MRA), which enables the representation of a function at different levels of resolution. By capturing local and global information, wavelets provide a flexible tool for tasks ranging from function approximation to data compression and reconstruction from samples [1]. This versatility has revolutionized the analysis of functions and signals in theoretical mathematics and engineering applications. The origins of wavelets date back to 1909 when Alfred Haar introduced the first wavelet basis [30], long before the term "wavelet" was coined. The Haar wavelet, while simple and orthogonal, is discontinuous, limiting its applicability in some practical contexts. However, Haar wavelets have proven useful in various theoretical contexts [29].

Significant advances in wavelet theory were achieved in the 1980s, notably by Yves Meyer [31] and Ingrid Daubechies [16]. Meyer's wavelets are constructed using a band-limited function, ensuring smoothness and regularity in the time domain. Although they lack compact support, their smoothness and orthogonality make them suitable for applications requiring continuously differentiable wavelets. In contrast, Daubechies wavelets are constructed to possess compact support and a specified number of vanishing moments. These properties enhance localization and facilitate the approximation of smooth functions. A key advantage of Daubechies wavelets is their ability to combine compact support with adjustable vanishing moments, making them particularly effective for smooth function approximation, signal analysis, and image processing [16, 39, 38]. The vanishing moments of the Daubechies wavelets play a crucial role in the multifractal formalism. More precisely, they make the wavelets orthogonal to low-order polynomials, enabling the removal of such trends. Sequentially, wavelet transforms based on Daubechies wavelets characterize Hölder spaces and pointwise regularity. They allow estimating the Hölder exponent through decay conditions in wavelet leaders [3].

Yves Meyer's pioneering work on wavelets laid the foundation for Stéphane Mallat's multiresolution analysis (MRA) in the scalar case. MRA provides a systematic and theoretical framework for constructing wavelet bases for spaces such as $L^2(\mathbb{R})$. A key starting point in MRA is to find a scaling function (father wavelet) that satisfies specific properties, from which the corresponding wavelet function (mother wavelet) is derived [31]. This framework enables the decomposition of signals across multiple scales, facilitating analysis at various levels of resolution, and permitting the construction of a wavelet basis. MRA also provides a theoretical framework for constructing multivariate scalar wavelet bases in spaces such as $L^2(\mathbb{R}^d, \mathbb{R})$. However, despite the general scheme of MRA, building such multivariate wavelet bases in practice remains challenging [27]. The primary difficulty lies in finding a suitable multivariate scaling